

Sparks in detector 7, and how they compare to detector 3 (June cosmic bench — g_det7_long)

MX17 cosmic-bench QA

July 5, 2026

1 Purpose

The det3 spark study (companion note `det3_spark_analysis`) found sparks that are Poisson in time, predominantly muon-induced, edge-localised, and neither a single flash nor a propagating discharge. det7 is the most spark-prone detector of the June fleet (38.9 % of firing events spark, vs 9.1 % for det3) and runs at a different operating point (drift 700 V, resist past optimum). We repeat the identical analysis on `mx17_det6_det7_overnight_6-26-26 / long_run` (det7 = FEU6/8, 3.84 h, **22 247 sparks**) to ask: is it the *same* phenomenon, just more of it?

2 Answer: yes — the same story, amplified

Every det3 conclusion reproduces in det7, scaled up by the higher operating gain (Table 1). Sparks are muon-seeded avalanche breakdowns whose probability and spatial/temporal extent grow as the detector is pushed closer to breakdown, initiated at the same edge / hot-strip weak spots, developing as a spatially-extended, temporally-spread event with no propagating front.

	det3 (g_det3_wknd)	det7 (g_det7_long)	reading
spark rate (of firing events)	9.1 %	38.9 %	det7 deep in sparking regime
mean rate	0.33 Hz	1.61 Hz	higher Poisson mean
random in time?	Poisson	Poisson	neither bursts (no HV instab.)
spark rate, muon <i>crosses</i> det	4.9 %	31.0 %	muon-induced
spark rate, muon <i>misses</i> (>30mm)	1.2 %	5.3 %	(stronger in det7)
crossing/miss enhancement	4.1×	5.9×	muon-dominated in both
spontaneous floor (miss rate)	1.2 %	5.3 %	floor rises with gain
sparks with no clean M3 track	60 %	47 %	DAQ cross-talk in both
strips/spark (X, Y)	40, 53	64, 76	bigger discharges; Y>X in both
spatial span (median)	~200 mm (half)	~360 mm (whole)	det7 fills the plane
occupancy peaks at	edges + hot strips	edges + hot strips	same weak spots
hit-time spread (median)	253 ns	334 ns	longer in det7
fraction $\sigma_t > 300$ ns	24 %	66 %	det7 rings longer
position-time corr $ r $	0.12	0.15	no “walk” in either

Table 1: det3 vs det7 — identical phenomenology, det7 simply further into breakdown.

3 Q1 — Time: Poisson, faster

det7 sparks at a flat 1.61 Hz with no bursts; inter-spark intervals are exponential and counts-per-bin have $\text{std} = \sqrt{\text{mean}}$ (Fig. 1) — Poisson, exactly as det3, just a 5× higher mean. The elevated rate is therefore *not* a new instability; it is the same random process with more muons reaching breakdown.

(Mild consecutive-event clustering, $P(\text{spark} | \text{prev spark}) = 39\%$ vs 32% baseline, matches det3's small excess.)

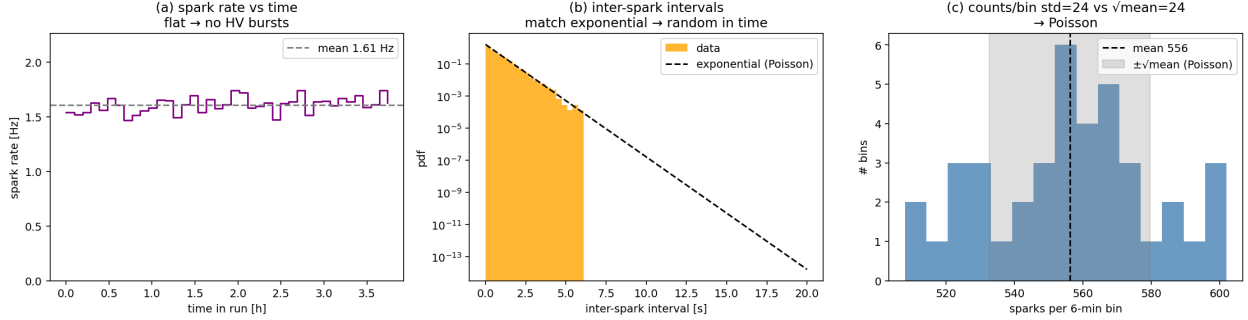


Figure 1: det7 time structure: flat rate (a), exponential intervals (b), Poisson counts/bin (c).

4 Q2 — Muon-induced, even more so

Conditioning on a clean single M3 muon, a track crossing det7 sparks at **31.0 %** versus **5.3 %** for one that clearly misses — a **5.9×** enhancement (Fig. 2a), stronger than det3's $4.1\times$. So $\sim 83\%$ of in-detector sparks are muon-induced. The spontaneous floor is higher than det3 (5.3% vs 1.2%), consistent with running closer to breakdown where even a stray electron can trigger a discharge. The discharge centroid is again central and compact (Fig. 2c). The M3 crossing map (Fig. 2b) shows more structure than det3's — but this tracks the overall M3 track density / telescope acceptance, not a det7 hot-spot; the detector-frame centroid is the clean spatial measure. The “no clean track” population persists (47% of sparks vs 35% baseline; same $1.3\times$ over-representation and same discharge size), the DAQ common-mode cross-talk flagged for det3.

Shower hypothesis ruled out (as in det3). If the no-track sparks were multi-particle showers they would leave M3 with ≥ 2 track candidates; instead only **0.5 %** of det7 no-track sparks do (they are 62% M3-blank, 38% single degraded track), and their discharge morphology is *identical* to muon-tracked sparks (median 122 vs 125 strips, span 389 vs 392 mm, spread 335 vs 333 ns; Fig. 3). A spark nearly doubles M3's no-track rate ($18 \rightarrow 29\%$ blank) — cross-talk, not showers. This reproduces the det3 result on a $2.5\times$ larger spark sample.

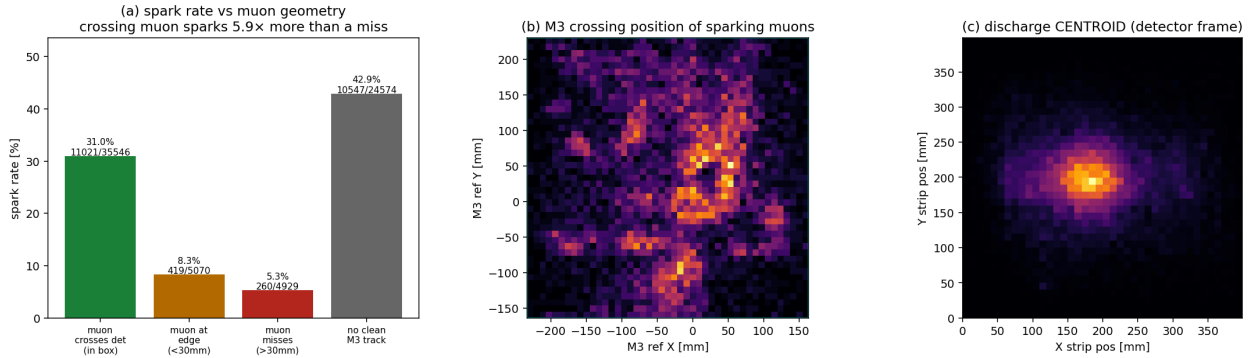


Figure 2: det7 muon dependence: (a) crossing muons spark $5.9\times$ more than missing ones; (b) M3 crossing positions; (c) central discharge centroid.

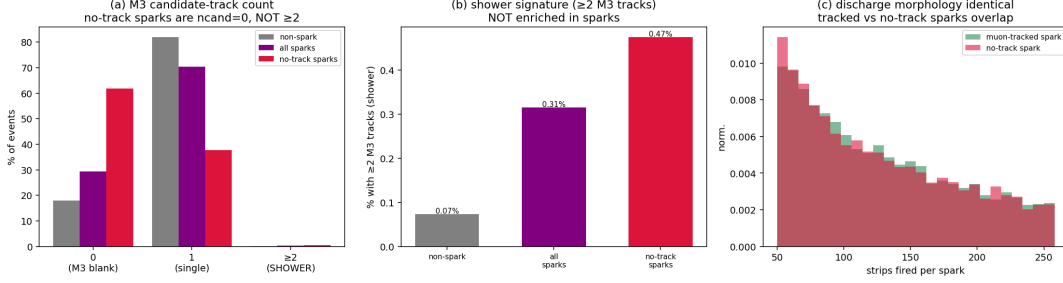


Figure 3: det7 shower test: (a) M3 candidate-track count by spark class; (b) the ≥ 2 -track shower fraction stays at the $\sim 0.5\%$ level; (c) no-track and muon-tracked sparks have identical discharge size — showers ruled out.

5 Q3 — Same places: edges and hot strips

det7 strip occupancy peaks at the plane **edges** (both planes; strongest at low- X and high- Y) and on a few **hot strips** — most dramatically an X/FEU6 channel near 45 mm that fires in a large fraction of all sparks (Fig. 4a), plus fixed Y/FEU8 spikes (Fig. 4b). This is the same edge+hot-strip signature as det3, more pronounced. Discharges are larger (median 64 X / 76 Y strips) and now span essentially the *whole* plane (~ 360 mm of 399 mm), versus half the plane in det3 — the higher gain lets a discharge propagate across the full active area. The Y/FEU8 plane again discharges harder than X, consistent with det7’s known FEU8 saturation band. Only 14 % of spark hits saturate the ADC.

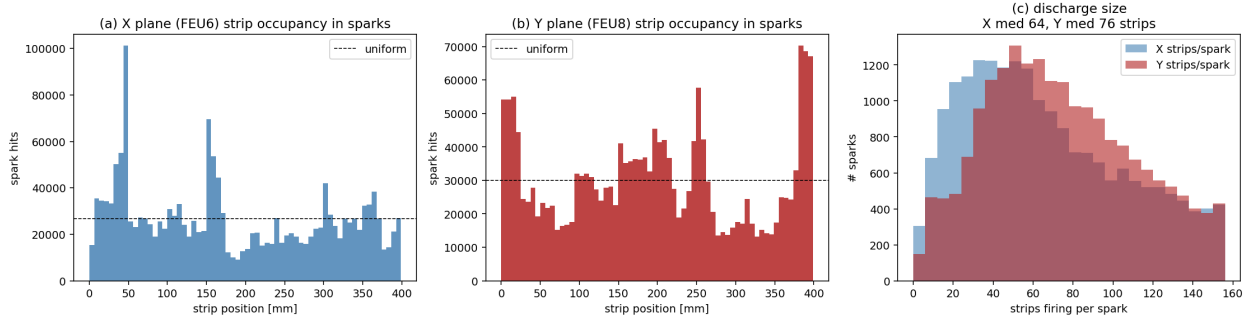


Figure 4: det7 locations: (a,b) edge peaks + hot strips in X and Y; (c) discharge size — larger than det3, Y harder than X.

6 Q4 — Still no “walk”, but longer-lived

det7 sparks are, like det3, neither simultaneous (0.8 % within 100 ns) nor propagating: the position–time correlation is $|r| \approx 0.15$ and the spark-centred Δpos – Δtime map (Fig. 5b) is again a cross with no tilt. The difference is duration: the median per-spark time spread is 334 ns (vs det3’s 253 ns) and **66 %** of det7 sparks spread beyond 300 ns (vs 24 % for det3) — the discharge rings/afterpulses for longer, filling much of the readout window (Fig. 5a,c). Same mechanism (extended discharge + afterpulsing, timing independent of position), longer tail.

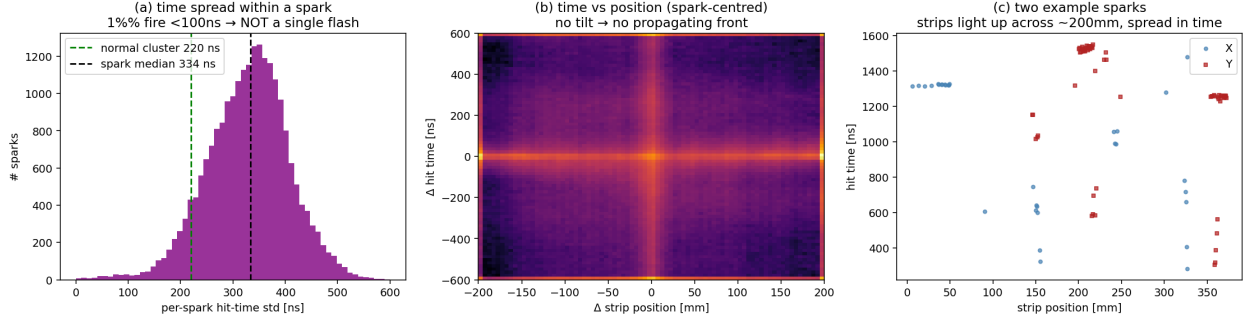


Figure 5: det7 time structure: (a) longer per-spark spread than det3; (b) no tilt (no propagating front); (c) example sparks across the whole plane, spread in time.

7 Takeaways

- det7 and det3 spark by the **same mechanism**; det7 is simply operated further into breakdown, which raises the rate ($0.33 \rightarrow 1.61$ Hz), the muon enhancement ($4.1 \rightarrow 5.9 \times$), the spontaneous floor ($1.2 \rightarrow 5.3$ %), the discharge size (half $\rightarrow$ whole plane) and its duration ($253 \rightarrow 334$ ns).
- The lever is the **operating gain**: det7 needs a lower resist HV (the HV-scan optimum was ~ 440 V; this run sat above it). Lowering it cuts the rate but not the muon-seeded nature.
- The **edge / hot-strip** weak spots and the **DAQ common-mode cross-talk** (spark $\rightarrow$ M3 track lost) reproduce on a second detector, so both are systemic, not det3 quirks — worth fixing at the detector-edge/guard-ring and DAQ level.

Reproduce: `extract_sparks.py g_det7_long` then `make_plots.py g_det7_long`; numbers in `spark_meta.json`. Identical pipeline to `det3_spark_analysis`.